

SkillPulse

Project Status Report

Done vs. Not Done - Compared to Project Overview PDF

Generated: July 19, 2026

Repository: skillpulse (v7.0)

This report compares the current SkillPulse codebase against the original Project Overview & Build Plan (SkillPulse_Project_Overview.pdf). Overall: the core pipeline is built and working, but several items from the plan - weekly automation, richer features, cloud deployment, and documentation polish - are not yet complete.

Executive Summary

COMPLETED (~65%): End-to-end pipeline from Adzuna scraping through skill extraction, XGBoost training, FastAPI serving, Streamlit dashboard, Evidently drift monitoring, conditional retraining, and Docker Compose.

NOT COMPLETED (~35%): Weekly re-scraping, RemoteOK integration, spaCy/LLM benchmark story, time-based splits, linear regression baseline, seniority/location/recency features, 'skills to learn next' recommender, cloud deployment, case-study README, and demo video.

What Is DONE

- [+] Data ingestion via Adzuna API (India, UK, US) [notebooks/01_setup_and_scrape.ipynb]
- [+] 4 search keywords scraped (data scientist, SWE, MLE, backend dev) [01_setup_and_scrape.ipynb]
- [+] ~6,000 job postings collected (2,000 per country) [MySQL job_postings table]
- [+] DB-first MySQL architecture (5 tables) [schema_mysql.sql + notebooks]
- [+] HTML text cleaning and normalization [02_cleaning_and_extraction.ipynb]
- [+] Fingerprint deduplication (~62% duplicate reduction) [02_cleaning_and_extraction.ipynb]
- [+] Regex-based skill extraction (~46 skills) [02_cleaning_and_extraction.ipynb]
- [+] Skill-to-job many-to-many mapping in MySQL [skills + job_skills tables]
- [+] EDA: salary distributions, currency standardization (USD) [03_eda_and_feature_engineering.ipynb]
- [+] Log1p target transformation for regression [03_eda_and_feature_engineering.ipynb]
- [+] India excluded from salary training (documented trade-off) [03 notebook + README]
- [+] Feature matrix exported (48 features: 46 skills + 2 country flags) [data_exports/]
- [+] XGBoost model with Optuna hyperparameter tuning (30 trials) [04_model_training.ipynb]
- [+] 5-fold cross-validation during tuning [04_model_training.ipynb]
- [+] Model saved as joblib [models/xgboost_model.joblib]
- [+] Model run logged to MySQL [model_runs table]
- [+] Test MAE ~0.31 (log scale), ~\$31k USD average error [04_model_training.ipynb]
- [+] FastAPI: GET /health [app/main.py]
- [+] FastAPI: GET /skills/top and /skills/top/{country} [app/main.py]
- [+] FastAPI: POST /predict (GB + US only) [app/main.py]
- [+] Streamlit dashboard - 4 views [streamlit_app/app.py]
- [+] Market Overview (stats, country pie, salary coverage)
- [+] Skill Intelligence (top skills, co-occurrence, high-value skills)
- [+] Salary Predictor (multi-select skills, gauge, matching jobs)
- [+] Model Diagnostics (run history, params, feature importance)
- [+] Evidently AI drift monitoring [mlops/drift_monitor.py]
- [+] Auto-retrain on drift >= 30% [mlops/retrain.py]
- [+] Docker Compose (MySQL + API + Streamlit + drift profile) [docker-compose.yml]
- [+] Windows control panel (start services, drift, retrain) [start_services.bat]
- [+] Technical README and AI_MEMORY design doc [README.md, AI_MEMORY.md]
- [+] EDA plots (salary dist, skills, co-occurrence, feature importance) [eda_plots/]

What Is PARTIALLY Done

- [~] Skill extraction NLP: Regex only - spaCy mentioned in overview but not implemented
- [~] Baseline model comparison: Default XGBoost used as baseline, not Linear Regression
- [~] Train/test split: Random 80/20 split - overview requires time-based split
- [~] Feature engineering: Skills + country only - no seniority, location, or recency
- [~] MLOps automation: Drift + retrain scripts exist but no Airflow/cron scheduler
- [~] Deployment: Dockerized locally - not deployed to Render or Hugging Face
- [~] Streamlit <-> FastAPI integration: Both exist but Streamlit loads model/DB directly, not via API
- [~] Documentation: Strong technical README - not written as case study with surprising findings
- [~] Database schema file: schema_mysql.sql missing 'country' column used everywhere in code
- [~] Data volume: 6,000 postings (one-time) vs 8,000+ weekly in overview

What Is NOT Done

- [-] RemoteOK API integration (only Adzuna implemented)
- [-] Weekly automated re-scraping of job postings
- [-] Airflow or cron-based scheduling for retrain pipeline
- [-] spaCy NLP for skill extraction
- [-] BeautifulSoup for HTML parsing (regex used instead)
- [-] LLM-based NER skill extractor
- [-] LLM vs regex benchmark with cost/accuracy documentation
- [-] Time-based train/test split (train on older, test on newest)
- [-] Linear Regression baseline model
- [-] Documented LR vs XGBoost comparison (% improvement)
- [-] Seniority level feature extraction
- [-] Location / fine-grained geo features
- [-] Days-since-posted / recency features
- [-] 'Skills to learn next' personalized recommender in GUI
- [-] Streamlit calling FastAPI backend (currently standalone)
- [-] Cloud deployment to Render or Hugging Face Spaces
- [-] Public live demo URL
- [-] README 'Surprising Findings' section
- [-] README 'What Didn't Work' section (e.g. LLM-NER dropped)
- [-] 60-90 second demo video/GIF embedded in README
- [-] Resume bullet placeholders filled (LR improvement %, surprising skill insight)
- [-] Standalone adzuna_scraper.py script (scraping is notebook-only)

11-Step Pipeline Scorecard

Step	Status	Notes
1. Data collection / EDA	Partial	Adzuna only; EDA done; no RemoteOK
2. Data processing	Done	Clean, dedupe, missing salaries handled
3. Feature engineering	Partial	Skills only; no seniority/location/recency
4. Train/test split	Not Done	Random split, not time-based
5. Train model	Partial	XGBoost only; no Linear Regression baseline
6. Evaluation	Partial	XGBoost metrics logged; no LR comparison
7. Return / select model	Done	XGBoost saved with joblib
8. Dump model + FastAPI	Done	API live with predict endpoint
9. Streamlit GUI	Partial	Salary yes; 'learn next' recommender no
10. Deployment	Partial	Docker only; no Render/HF deployment
11. MLOps (weekly + drift)	Partial	Evidently + retrain yes; weekly automation no

Resume Claims - Can vs Cannot Say Today

Safe to claim:

- [+] Built end-to-end MLOps pipeline scraping Adzuna across 3 countries
- [+] Regex skill extraction with 46 tech skills mapped to job postings
- [+] XGBoost salary predictor with Optuna tuning (MAE ~\$31k USD)
- [+] FastAPI REST API + Streamlit analytics dashboard
- [+] Docker Compose deployment with Evidently drift monitoring
- [+] Documented engineering trade-off: India excluded from salary training

Do NOT claim yet (not implemented):

- [-] 8,000+ job postings scraped weekly
- [-] spaCy / LLM-based NER with benchmark documentation
- [-] Outperformed linear regression baseline by X%
- [-] Automated weekly retraining via Airflow
- [-] Deployed live Streamlit app at public URL
- [-] GUI recommends which skills to learn next

Recommended Next Steps (Priority Order)

- [~] [P1 - Critical] Fix schema_mysql.sql: add missing 'country' column to job_postings
- [~] [P1 - Critical] Add Linear Regression baseline + document XGBoost improvement %
- [~] [P2 - High] Switch to time-based train/test split
- [~] [P2 - High] Add 'skills to learn next' recommender to Streamlit Salary Predictor
- [~] [P2 - High] Add seniority + recency features to feature engineering
- [~] [P3 - Medium] Wire Streamlit to FastAPI (or document architectural choice)
- [~] [P3 - Medium] Add cron/Airflow stub for weekly drift check + retrain
- [~] [P3 - Medium] Rewrite README as case study with surprising findings
- [~] [P4 - Polish] Deploy to Render or Hugging Face Spaces
- [~] [P4 - Polish] Record 60-90s demo video for README
- [~] [P4 - Polish] Document LLM vs regex experiment (even if regex wins)
- [~] [P4 - Polish] Remove stray notebooks/3.6 pip log file

Report generated automatically from codebase audit against SkillPulse_Project_Overview.pdf. For technical details see README.md and AI_MEMORY.md.